



DEALER BEST PRACTICE SERIES

Process for Removing Rust from Fuel Tanks

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1.0 Introduction

A new process to remove rust from a fuel tank has been implemented at Machining Workshop in Ferreyros S.A using a chemical rust solvent instead of sandblasting. The process reduces cost to clean fuels tanks by 70%.

2.0 Best Practice Description

The focus of this Best Practice is a new method to remove the rust from a fuel tank by applying a chemical solvent (see the picture 1). Solvent selected does not have any environmental or safety hazards, is very effective, and can be reused. This operation takes two hours for each side of fuel tank and can be performed by one technician.

Dealer uses Evapo-rust. Evapo-rust information and specifications is ready available on internet and product available through multiple sources.



Picture 1. Evapo-rust barrel liquid (55 gallons)

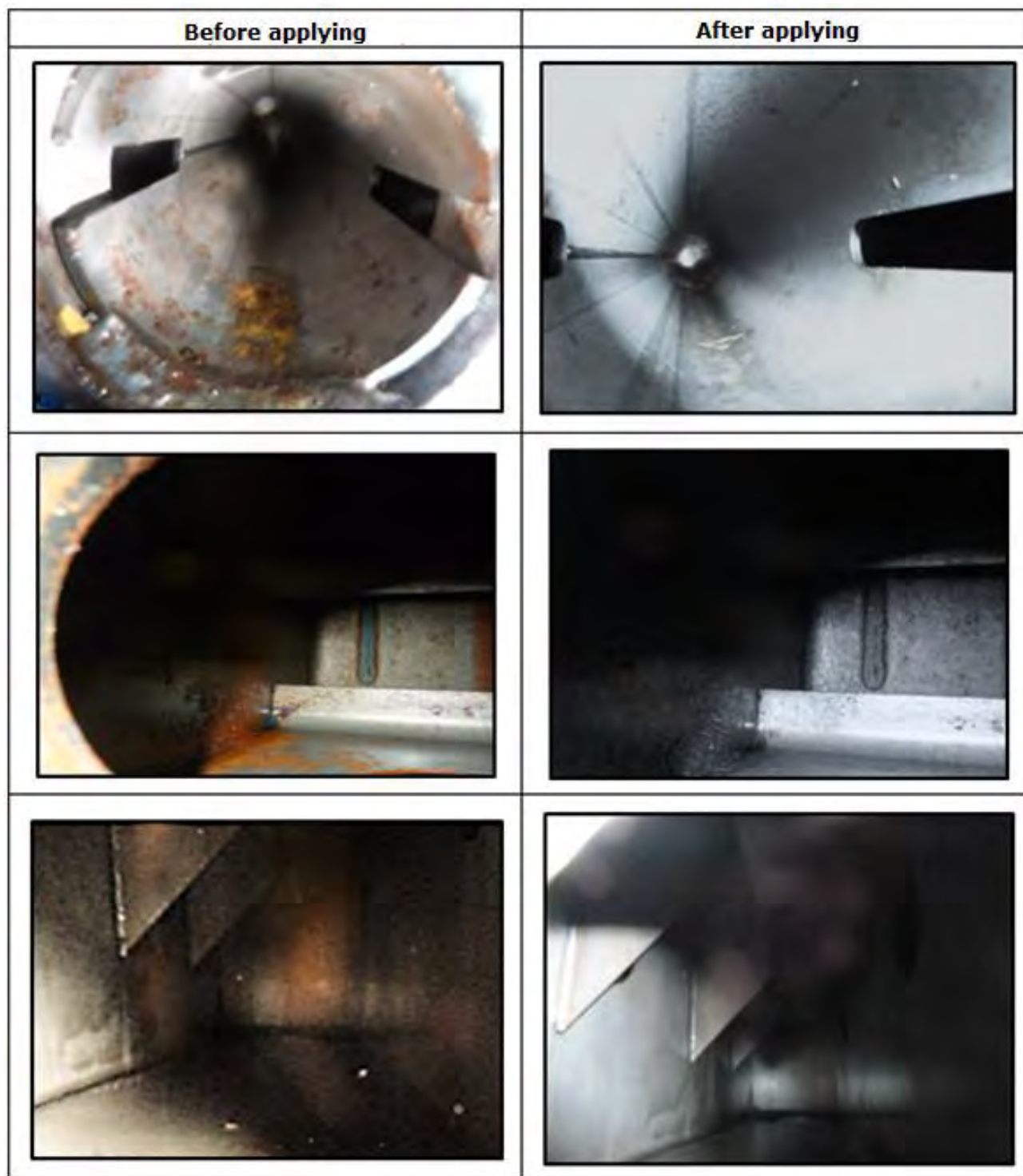
Our former process to clean rust from fuel tanks was to cut a plate out of the fuel tank for access and sandblasting the inside of the tank to remove rust. The process required up to four technicians (two welders, two handlers) and took a long time (five days approximately).

With the new process using solvent, time to clean the fuel tanks was reduced from five days down to 1.5 days. Costs include the cost of the solvent was reduced from \$6500 to \$818.

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The results after application of Evapo-rust inside the fuel tank are shown in the picture 2.



Picture 2. Before

After

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3.0 Implementation Steps

Process

- Check rust level inside the fuel tank through its holes (See the picture 3).



Picture 3. Inside the fuel tank

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- It is not necessary to completely fill the tank with solvent. With only a partial fill, the tank can be rotated in order to clean all the rust inside the tank. Two hours in each position is sufficient for cleaning the rust. For example, in the following pictures the fuel tank 785C was cleaned using 110 gallons of solvent. Solvent can be reused.



Fill the fuel tank and first position



Second position



Flip position



Third position

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Fourth position



Fifth position



Sixth position



Drain the Evapo-rust liquid

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- Clean inside the fuel tank with cold water.



Picture 4. Cleaning with cold water

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- The fuel tank must be completely dry inside. Apply VCI (Vapor Corrosion Inhibitors) after that.



Picture 5. Application of VCI

- Finally, the fuel tank is wrapped with Stretch Film.



Picture 5. Wrapped tank with Stretch Film

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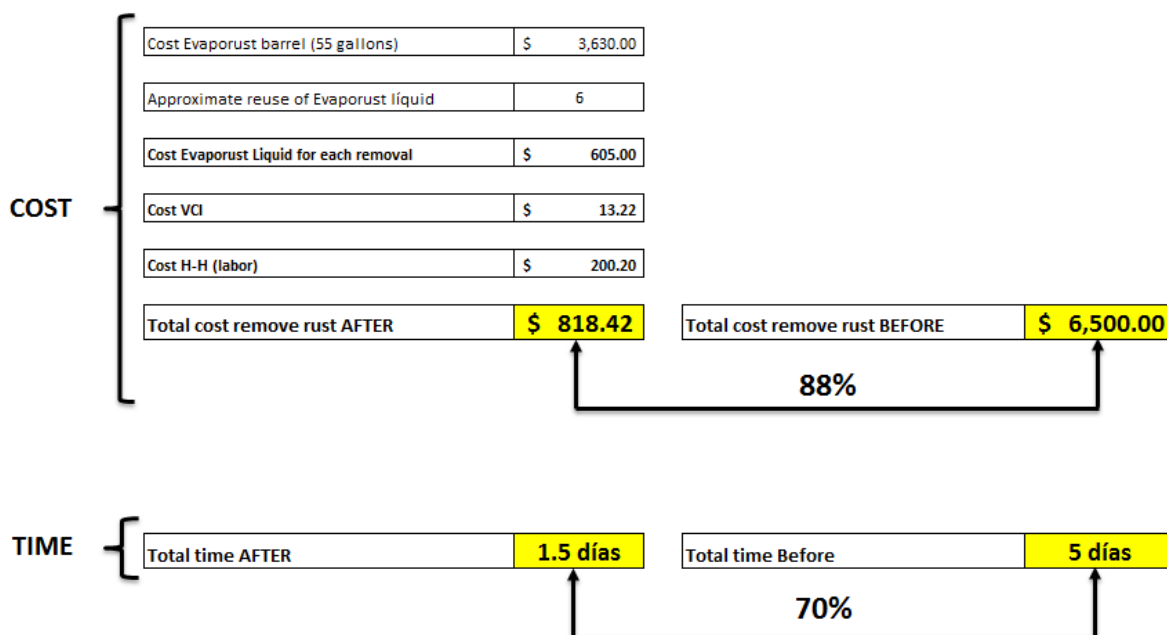
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4.0 Benefits

- Reduce the use of labor 80%.
- Excellent rust removal
- No messy residues to remove
- Reduce the turnaround time to remove rust in 70% by each fuel tank (see the picture 6).
- Reduce the cost to remove rust in 88% by each fuel tank (see the picture 6).
- Evapo-rust has multiple other rust removal applications.



Picture 6. Comparison between cost and time – after and before this best practice

5.0 Resources Required

- See the picture 6.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

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8.0 Acknowledgements

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